

Assignment for March 16, 2026

NIST AI RISK
MANAGEMENT
FRAMEWORK



Ethical Frameworks and Cybersecurity

Isaac Adebayo

Monday, March 16, 2026

MSDS 570-01

Group 4: (Work of Lisette Batiste)

Name of Ethical Framework

NIST AI Risk Management Framework (AI RMF)

Summary of the Ethical Framework

The NIST Artificial Intelligence Risk Management Framework (AI RMF) was developed by the National Institute of Standards and Technology to guide organizations in **designing, developing, and deploying trustworthy AI systems**. The framework focuses on **managing risks related to AI technologies while promoting responsible and ethical use**.

According to NIST (2023), the framework identifies several **key characteristics of trustworthy AI, including validity, reliability, safety, security, accountability, transparency, explainability, fairness, and privacy protection**. These principles are intended to ensure that AI systems operate in ways that are technically sound and socially responsible.

The AI RMF is organized around four core functions: **Govern, Map, Measure, and Manage**. These functions help organizations **evaluate risks, implement safeguards, and continuously monitor AI systems** to ensure ethical and secure outcomes.

Reference: National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0).



How it Applies to Cybersecurity

The NIST AI Risk Management Framework directly supports cybersecurity by **emphasizing security, resilience, and risk management in AI systems**. Many modern cybersecurity tools rely on machine learning models to detect threats, analyze network activity, and identify malicious behavior. However, **these systems can also introduce new vulnerabilities if they are not properly designed or monitored.**

For example, AI systems used in cybersecurity can be **vulnerable to adversarial attacks, where malicious actors manipulate data to deceive machine learning models**. The AI RMF encourages organizations to identify these risks early and implement safeguards such as model testing, monitoring, and validation.

The framework also highlights the importance **of accountability and transparency in AI-driven cybersecurity** systems. Security teams must understand how AI models make decisions so they can respond appropriately when threats occur. By integrating ethical risk management into cybersecurity practices, organizations can improve both the effectiveness and trustworthiness of their security systems.

How does the ethical framework focus on cybersecurity in a Data Science Workflow?

The NIST AI Risk Management Framework can **be integrated into the data science workflow to ensure that AI models are developed responsibly and securely.** A typical data science workflow includes stages such as data collection, data preparation, model development, evaluation, and deployment. Each of these stages presents potential ethical and security risks.

During data collection, **organizations must ensure that data is obtained legally and ethically while protecting individual privacy.** During data preparation and modeling, developers should evaluate datasets for bias and ensure that models produce fair and reliable outcomes.

In the deployment phase, cybersecurity practices become especially important. **Models must be protected against attacks, unauthorized access, and data leaks.** Continuous monitoring and risk assessment are necessary to ensure that AI systems remain secure and perform as intended.

Applying the AI RMF to the data science lifecycle helps organizations build systems that are secure, transparent, and ethically responsible.

What is the related Article, Journal Paper, Conference paper

Article:

Brundage, M., et al. (2018). The Malicious Use of Artificial Intelligence: Forecasting, Prevention, and Mitigation. Future of Humanity Institute, University of Oxford.

Summary:

This paper examines how artificial intelligence technologies can be exploited for harmful purposes such as automated cyberattacks, misinformation campaigns, and surveillance. The authors argue that organizations must implement stronger risk management strategies to prevent the misuse of AI systems. The study highlights the importance of ethical frameworks like the NIST AI Risk Management Framework to ensure that AI technologies are developed with appropriate safeguards. By identifying potential risks early and implementing security controls, organizations can reduce the likelihood that AI systems will be used maliciously.

